

Math 520, F15

Quiz 11

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Use Laplace transform to solve the initial value problem:

$$y'' + 25y = 0, \quad y(0) = 4, \quad y'(0) = 5$$

Solution: Let $Y(s) = \mathcal{L}\{y(t)\}$, then

$$\mathcal{L}\{y'(t)\} = sY(s) - y(0) = sY(s) - 4$$

$$\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0) = s^2Y(s) - 4s - 5$$

$$\Rightarrow \mathcal{L}\{y''\} + 25\mathcal{L}\{y\} = \mathcal{L}\{0\}$$

$$[s^2Y(s) - 4s - 5] + [25Y(s)] = 0$$

$$(s^2 + 25)Y(s) = 4s + 5$$

$$Y(s) = \frac{4s + 5}{s^2 + 25}$$

$$\frac{4s + 5}{s^2 + 25} = \frac{4s + 5}{s^2 + 5^2} = 4 \frac{s}{s^2 + 5^2} + \frac{5}{s^2 + 5^2}$$

$$\Rightarrow y(t) = \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{4 \frac{s}{s^2 + 5^2} + \frac{5}{s^2 + 5^2}\right\}$$

$$= 4 \mathcal{L}^{-1}\left\{\frac{s}{s^2 + 5^2}\right\} + \mathcal{L}^{-1}\left\{\frac{5}{s^2 + 5^2}\right\} = 4 \cos 5t + \sin 5t$$

Problem 2. (4 points) Transform the given equation into a system of first order equations:

$$y''' + ty' + (t^2 - 2)y = 3 \sin t. \quad \Rightarrow \quad y''' = -ty' - (t^2 - 2)y + 3 \sin t$$

Solution: Let $x_1 = y$, $x_2 = y'$, $x_3 = y''$

then

$$\begin{cases} x_1' = x_2 \\ x_2' = x_3 \end{cases}$$

$$x_3' = -tx_2 - (t^2 - 2)x_1 + 3 \sin t.$$